

PROJECT AND TEAM INFORMATION

Project Title

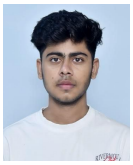
DBOS-Based Resource Management Platform with EV Charging as a Real-World Workload

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Project Code	OSDBMS-V-2026-T181
Team Name	HECODER11
Academic Year	2026
Department	Computer Science and Engineering

Team Information

Details of the project team members and their assigned roles.

S.No.	Photo	Name / Role	Student ID / Email
1		**Saif Hussain** *Team Lead*	**Student ID:** 240211783 **Email:** 240211783@geu.ac.in
2		**Adarsh Priyadarshi** *Member*	**Student ID:** 240211475 **Email:** 240211475@geu.ac.in
3		**Siddharth Singh Bisht** *Member*	**Student ID:** 240212225 **Email:** 240212225@geu.ac.in
4		**Bharat Gangwar** *Member*	**Student ID:** 240211932 **Email:** 240211932@geu.ac.in

Team Name: HECODER11
Project Code: OSDBMS-V-2026-T181

The team is working on the design and development of a DBOS-based resource management platform in which EV charging is used as a real-world workload for demonstrating operating-system scheduling, resource allocation, database persistence and concurrency management.

PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

Brief restatement of the project's main goal.

The project proposes a DBOS-based resource management platform that integrates Operating System scheduling and resource-allocation concepts with Database Management System capabilities. Electric vehicle charging is selected as a real-world workload because multiple EVs may compete for a limited number of charging resources. The system treats charging requests as workloads or jobs and charging stations as limited system resources. A scheduling component determines the order in which requests should be served, while a resource manager allocates available charging stations. The database maintains persistent information about users, electric vehicles, charging stations, requests, and charging sessions. Phase II focuses on converting the initial concept into a detailed technical design covering system architecture, database schema, scheduling, resource allocation, transaction handling, testing, and the implementation roadmap for Phase III.

Updated Project Approach and Architecture (2 pts)

Describe the current approach, including system design, database structure, scheduling strategy and implementation approach.

The proposed system follows a layered resource-management approach that integrates request management, scheduling, resource allocation, database management, EV charging resources, and monitoring and reporting. An EV user first submits a charging request, after which the request-management component validates the request and places it in the waiting queue. The scheduling engine evaluates pending requests using factors such as priority, waiting time, battery urgency, and charger compatibility to determine the appropriate execution order. The resource manager then checks the availability of compatible charging stations and reserves a suitable charger for the selected request. Once a charger is allocated, a charging session is created and the corresponding charger and request states are updated in the database. The database layer maintains persistent information about users, electric vehicles, charging stations, charging requests, and charging sessions, while transaction control is used during resource allocation to reduce the possibility of inconsistent states and double allocation. The architecture combines Operating System concepts such as scheduling, resource allocation, resource states, and workload management with DBMS concepts such as relational storage, primary and foreign keys, transactions, constraints, and concurrency control. This approach provides the technical foundation for implementing and evaluating the complete EV charging resource-management platform during Phase III.

Resource Management Model

Mapping Operating System resource-management concepts to the EV charging workload.

OS Concept	EV Charging Equivalent
Process / Job	EV charging request
Scheduler	Charging request scheduling engine
Resource	Charging station / charging point
Ready Queue	Waiting charging requests
Resource Allocation	Assignment of charger to EV
Process Completion	Completion of charging session
Resource Release	Charger becomes available
Process State	WAITING, ALLOCATED, CHARGING and COMPLETED states
Database State	Persistent record of EV, request, charger and session information
Concurrency Control	Prevention of double charger allocation

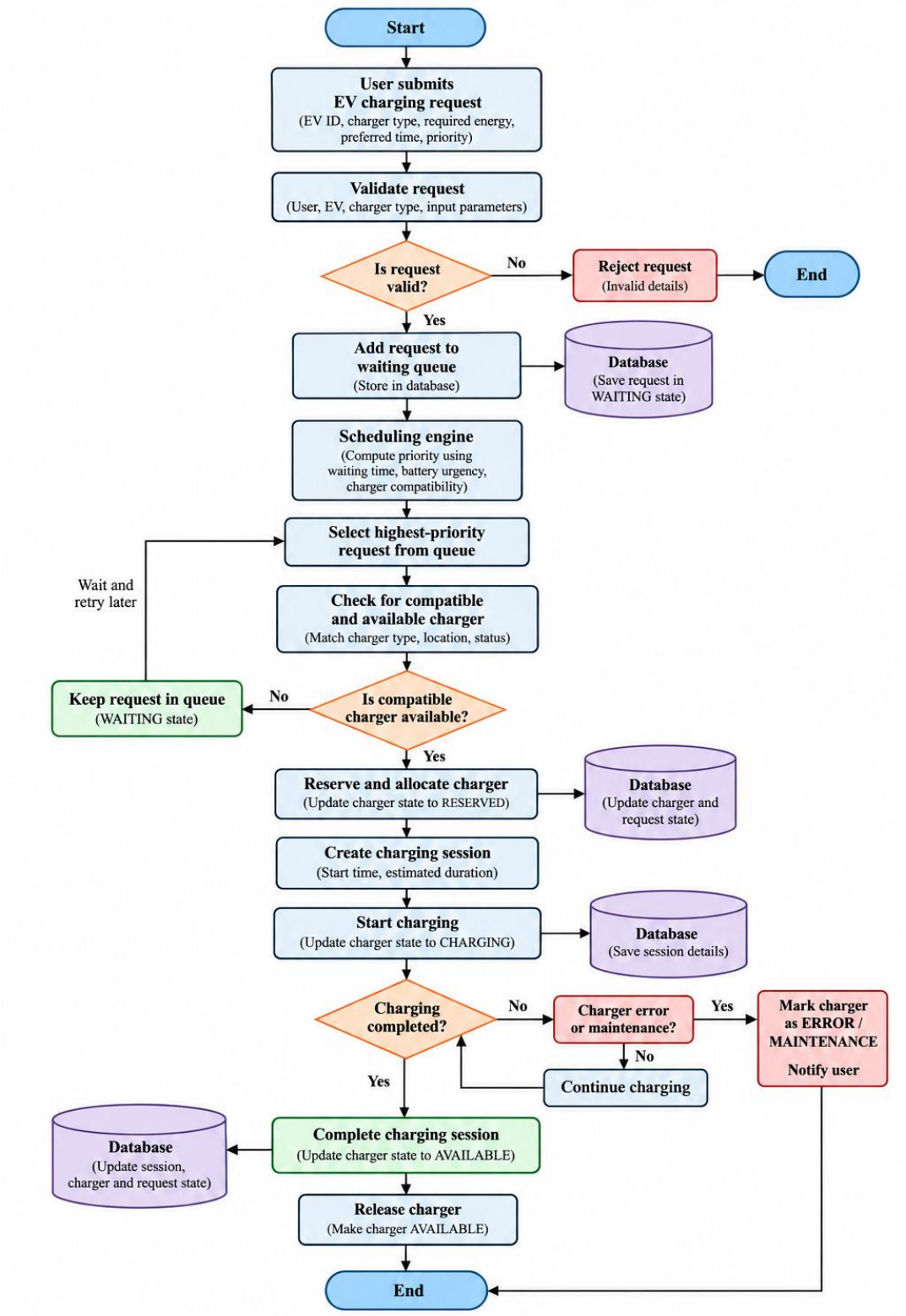
Charging Resource States

Logical states maintained by the charging resource manager.

State	Description
AVAILABLE	Charger is free and can be allocated.
RESERVED	Charger has been assigned to a request but charging has not started.
CHARGING	Charger is actively serving an EV.
COMPLETED	Charging operation has completed and the session is being finalized.
ERROR / MAINTENANCE	Charger is unavailable because of a fault or maintenance activity.

Resource Management Flowchart

Flow of an EV charging request from submission to resource release.



EV Charging Resource Management Workflow

Entity Relationship Diagram

Detailed database relationship model for the EV charging resource-management platform.

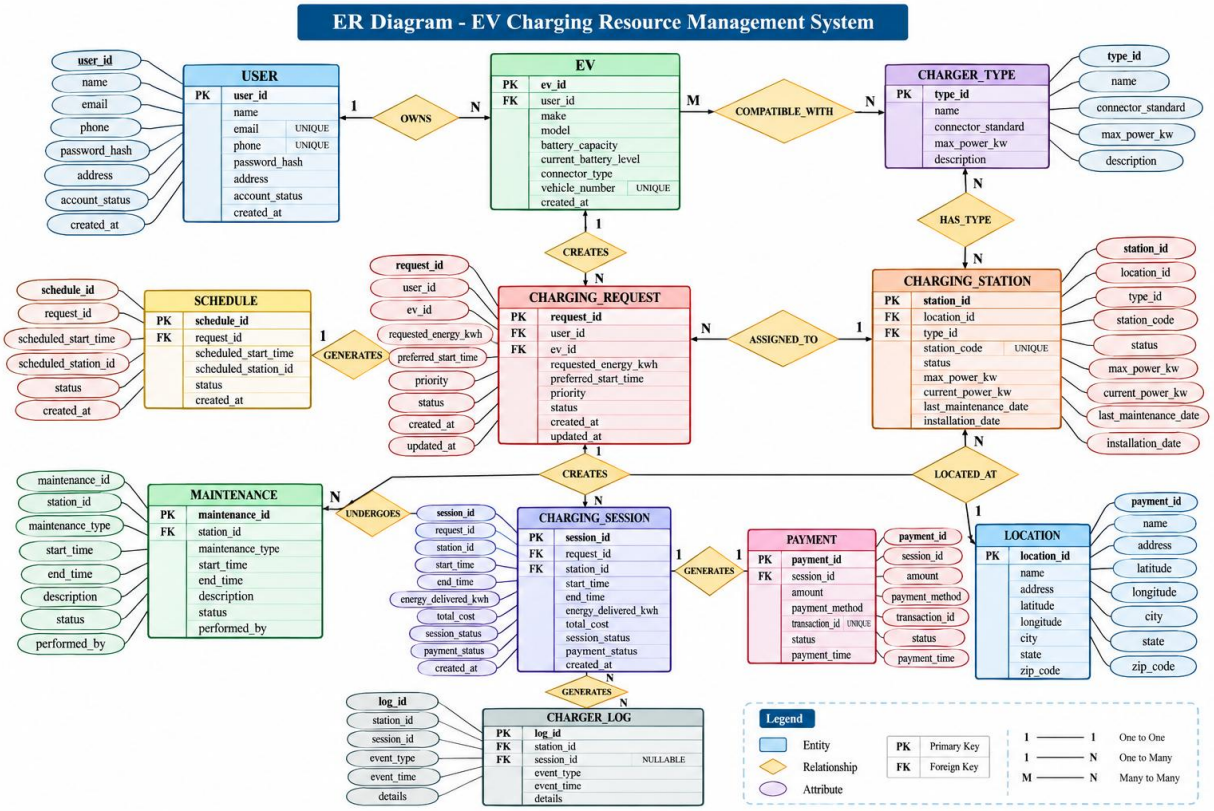


Figure: Entity Relationship Diagram of the EV Charging Resource Management System

Explanation of the ER Diagram

The ER diagram represents the database structure of the EV charging resource-management platform and illustrates the relationships between its major entities. The central entities include USER, EV, CHARGING REQUEST, CHARGING STATION and CHARGING SESSION. Supporting entities such as SCHEDULE, LOCATION, CHARGER TYPE, PAYMENT, MAINTENANCE and CHARGER LOG provide additional information related to scheduling, resource management, monitoring, payments, and maintenance activities.

A user can own multiple electric vehicles, and each EV can generate multiple charging requests. These requests are processed by the scheduling and resource-management system according to resource availability and scheduling requirements. When a suitable charging station is allocated, a charging session is created to record the charging operation and associate it with the relevant EV and charging station. This relationship helps the system track resource usage and manage the complete charging lifecycle.

The diagram also represents the relationships between charging stations, locations, and charger types. Location information identifies where charging resources are situated, while charger-type information helps determine compatibility between an EV and an available charging resource. The SCHEDULE entity supports the organization of charging requests, and PAYMENT records can store information associated with charging sessions. In addition, MAINTENANCE records help track servicing activities, while CHARGER LOG maintains operational information about charger status and events.

Primary keys are used to uniquely identify records in each entity, while foreign keys establish relationships between related entities and maintain referential integrity. The database design provides a structured way to store and retrieve information about users, EVs, charging requests, charging stations, charging sessions, and supporting operations. Overall, the ER diagram forms the database foundation of the platform and supports persistent data storage, charging-resource allocation, scheduling, session tracking, and consistent management of EV charging operations.

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Tasks Completed (7 pts)

Main tasks completed during Phase II.

Task Completed	Team Member
Problem definition and EV charging workload identification	Team
System architecture and component identification	Saif Hussain
Database entity and relational schema design	Adarsh Priyadarshi
ER model and database relationship design	Siddharth Singh Bisht
Scheduling and resource-allocation strategy	Bharat Gangwar
Charger state-transition model	Team
Transaction and concurrency-control design	Team
SQL DDL and sample database operations	Team
Testing and validation strategy	Team

Challenges/Roadblocks (7 pts)

Challenges faced during the design phase and proposed solutions.

The major challenge during Phase II was mapping real-world EV charging operations to Operating System resource-management concepts. This was addressed by treating charging requests as workloads or jobs and charging stations as limited resources. Another challenge was designing a database structure that could maintain the current state of charging resources while also preserving information about charging requests and sessions. This was addressed by defining separate entities for users, electric vehicles, charging stations, charging requests, and charging sessions with appropriate primary-key and foreign-key relationships. Concurrency was another important challenge because multiple EV requests may attempt to reserve the same charger at nearly the same time, which can lead to double allocation or inconsistent database states. The proposed solution is to use transaction-based resource allocation along with explicit charger states such as AVAILABLE, RESERVED, CHARGING, COMPLETED, and ERROR/MAINTENANCE. Designing an effective scheduling strategy was also a challenge because a simple first-come-first-served approach may not always provide efficient resource utilization. Therefore, the proposed scheduling approach considers request priority, waiting time, battery urgency, and charger compatibility. The remaining challenge is to integrate these designed components into a working prototype and validate their performance through functional, concurrency, and performance testing during Phase III.

Tasks Pending (7 pts)

Main implementation tasks remaining for Phase III.

Task Pending	Team Member
Implement the relational database and populate sample data	Team
Implement charging request module	Saif Hussain
Implement scheduling engine	Bharat Gangwar
Implement resource allocation and charger state management	Adarsh Priyadarshi
Integrate application and database	Siddharth Singh Bisht
Implement transaction and concurrency handling	Team
Execute functional and concurrency tests	Team
Measure waiting time, throughput and resource utilization	Team

Project Outcome/Deliverables (2 pts)

Key outcomes and deliverables of the project.

The expected outcome of the project is a working DBOS-based resource management platform that demonstrates the integration of Operating System and Database Management System concepts using EV charging as a real-world workload. The system is designed to manage multiple EV charging requests while efficiently allocating a limited number of charging resources. The major deliverables include a relational database for storing users, electric vehicles, charging stations, charging requests, and charging sessions; a charging request management module; a priority-based scheduling mechanism; a resource-allocation module for assigning and releasing chargers; a charger state-management mechanism; transaction and concurrency control for preventing double allocation and inconsistent database states; and testing and performance evaluation components. The completed system will demonstrate how scheduling, resource allocation, database persistence, transaction management, and concurrency control can work together to provide efficient and reliable management of EV charging resources.

Progress Overview (2 pts)

Summarize the current project progress, completed work and remaining work.

The project has completed the major Phase II design activities required to establish the technical foundation for the proposed EV charging resource-management platform. The problem domain and EV charging workload have been finalized, and the overall system approach has been defined. The database design has been prepared with entities and relationships for users, electric vehicles, charging stations, charging requests, and charging sessions, along with the required primary-key and foreign-key relationships. The scheduling strategy and resource-allocation workflow have also been designed, including the use of request priority, waiting time, battery urgency, and charger compatibility for scheduling decisions. The charger state model defines the major resource states, while transaction and concurrency strategies have been planned to reduce the possibility of double allocation and inconsistent database states. The ER diagram, resource-management flowchart, database structure, and testing strategy have also been prepared as part of the Phase II deliverables. The remaining work mainly involves implementing and integrating the designed components, executing functional and concurrency tests, and measuring performance parameters such as waiting time, throughput, and resource utilization during Phase III. Overall, Phase II has established the required design and planning foundation for developing and evaluating the working prototype.

Current Project Status

Component	Status	Remarks
Problem Definition	Completed	Workload and resource-management problem finalized.
System Architecture	Completed	Major system components and workflow designed.
Database Design	Completed	Entities, relationships and relational schema prepared.
Scheduling Strategy	Prepared	Priority-based scheduling approach defined.
Resource Allocation	Prepared	Allocation workflow and charger states defined.
Transaction Strategy	Prepared	Concurrency and rollback approach designed.
Implementation	Pending	Planned for Phase III.
Performance Evaluation	Pending	Waiting time and resource-utilization measurements planned.

Codebase Information (2 pts)

Repository link, branch and important implementation information.

Project Repository: https://github.com/saifhussain786/DBOS-Resource-Management-Platform
Primary Branch: main
Project Structure:

The planned codebase will contain separate modules for database connectivity, charging request management, scheduling, resource allocation, charger-state management, transaction handling and testing.

The database implementation will contain SQL DDL and sample data. The application layer will communicate with the relational database and execute scheduling and resource-allocation operations.

Important Components:

Database schema, charging request management, scheduling engine, resource manager, charger state management, transaction handling, testing and performance evaluation.

Testing and Validation Status (2 pts)

Provide information about tests planned or conducted during Phase II.

Test Type	Status	Notes
Database Schema Validation	Prepared	Primary and foreign-key relationships identified.
User and EV Registration	Planned	Will be tested during implementation.
Charging Request Creation	Planned	Request should enter the waiting queue.
Charger Allocation	Planned	Available compatible charger should be reserved.
Concurrent Allocation	Planned	Multiple requests must not allocate the same charger.
Transaction Rollback	Planned	Failed allocation should restore consistent state.
Charging Completion	Planned	Charger should return to AVAILABLE state.
Performance Testing	Pending	Waiting time, throughput and utilization will be measured.

Deliverables Progress (2 pts)

Current status of all major project deliverables.

Deliverable	Status
Project Problem Definition	Completed
System Architecture	Completed
Database Entity and Relationship Design	Completed
Relational Database Schema	Completed
Scheduling Strategy	Completed
Resource Allocation Design	Completed
Charger State Model	Completed
Transaction and Concurrency Design	Completed
SQL Implementation	In Progress
Application Integration	Pending
Functional Testing	Pending
Concurrency Testing	Pending
Performance Evaluation	Pending
Final Working Prototype	Pending